How-To Library

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Configuring Enterprise Data Preparation 10.2.2 on Microsoft Azure HDInsight

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Abstract

You can configure Informatica Enterprise Data Preparation to run in the cloud on Microsoft Azure. This article provides instructions and guidelines for configuring Enterprise Data Preparation to run on Microsoft Azure with an Azure HDInsight cluster.

Supported Versions

- Enterprise Data Preparation 10.2.2

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Overview

You can configure Enterprise Data Preparation to run on Microsoft Azure with an Azure HDInsight cluster.

You should have knowledge of Microsoft Azure and HDInsight. You should also have knowledge of deployment practices for Informatica Data Engineering Integration, Enterprise Data Catalog, and Enterprise Data Preparation.

For more information about Microsoft Azure, see the Microsoft product documentation.

To configure Enterprise Data Preparation on Microsoft Azure, perform the following steps:

1. Complete the prerequisites.
2. Set up the Data Preparation repository database in Microsoft Azure.
3. Configure data lake access in Microsoft Azure.
4. Install Enterprise Data Preparation in Microsoft Azure.
5. Use Informatica Administrator to create a Hadoop File System connection to each Azure Blob Storage and Azure Data Lake Storage location in the HDInsight cluster.

6. Use the Catalog Administrator to create Azure Blob Storage, Azure Data Lake Storage, and Hive resources in Enterprise Data Catalog.
   You must create a resource for each data storage location in the HDInsight cluster.

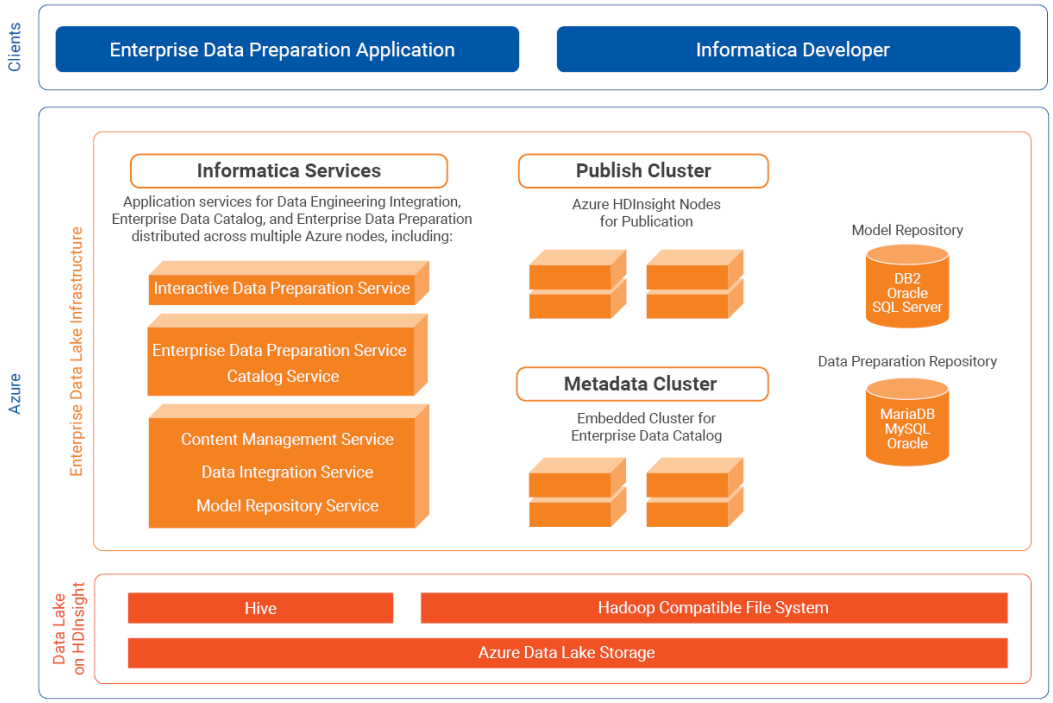
For more information about installing and configuring Informatica products, see the Informatica product documentation.

Enterprise Data Preparation on Microsoft Azure Architecture

When you deploy Enterprise Data Preparation on Azure, you install the Informatica domain, Enterprise Data Catalog, and Enterprise Data Preparation binaries on nodes running on Azure virtual machines. You also deploy the Model repository database and the Data Preparation repository database on Azure virtual machines.

Enterprise Data Preparation uses your data lake as deployed on Azure. The application goes through the Hadoop compatible file system to access data assets stored in one or more Azure Blob Storage containers or Azure Data Lake Storage Gen 1 locations in the HDInsight cluster. The application uses Hive to enable querying and analysis of assets available in the data lake.

The following figure shows the deployment architecture for configuring Enterprise Data Preparation on Azure:



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Prerequisites

Complete the following prerequisites before you deploy Enterprise Data Preparation on Microsoft Azure.

Enable Spark on the Azure HDInsight cluster.

The Data Integration Service associated with the Enterprise Data Preparation Service runs jobs on the Spark execution engine on HDInsight nodes. When you install the Spark package, you can set up an external shuffle service on each worker node in the cluster to dynamically adjust the resources Enterprise Data Preparation uses.

Install the Informatica services.

Install and configure the Informatica services in Microsoft Azure.

For information on installing the Informatica services, see the Informatica Data Engineering Installation and Configuration Guide.

Complete the Hadoop integration tasks for Azure HDInsight.

Complete the Hadoop integration steps described in the "Azure HDInsight Integration Tasks" section in the Informatica *Data Engineering Integration Guide*.

Install Enterprise Data Catalog.

Install and configure Enterprise Data Catalog in Microsoft Azure.

For information about installing Enterprise Data Catalog, see the *Informatica Enterprise Data Catalog Installation and Configuration Guide*.

Set Up the Data Preparation Repository Database

Set up the Data Preparation repository database on an Azure virtual machine. The Data Preparation repository stores worksheet metadata prepared by Enterprise Data Preparation users.

You can use a MySQL database, a MariaDB database, or an Oracle database as the Data Preparation repository database.

Allow 5 GB of disk space for the repository database. Allocate more space based on the amount of metadata you want to store.

MySQL and MariaDB Database Requirements

Perform the following steps to use a MySQL database or a MariaDB database as the Data Preparation repository.

Set the following system variables on the database server:

- For MySQL version 5.6.26 and higher, set lower_case_table_names=1.
- For MySQL version 5.7 and higher, set explicit_defaults_for_timestamp=1.

Ensure that the MySQL or MariaDB database has the following permissions:

- Create tables and views.
- Drop tables and views.
- Insert, update, and delete data.

The MySQL connector .jar file is not included with the installer. You must download the file and copy it to the following directory before you start the installer:

```
$USER_INSTALL_DIR$/services/shared/jars/thirdparty/
```

Make sure the name of the file is in the following format:

```
mysql-connector-java-<versiondetails>.jar
```

You must also set the mysql_connector_jar_path environment variable to the location of the MySQL connector .jar file when you run the installer to install Enterprise Data Preparation.

Oracle Database Requirements

Perform the following steps to use an Oracle database as the Data Preparation repository.

Ensure that the Oracle database has the following permissions:

- Create tables and views.
- Create sequence, session, and synonyms.
- Drop tables and views.
- Insert, update, and delete data.

Configure Data Lake Access

Before you install Enterprise Data Preparation, configure the access control lists (ACLs) required to access data in Azure Data Lake Storage locations in the HDInsight cluster.

Azure Data Lake Storage implements an access control model that derives from HDFS, which in turn derives from the POSIX access control model.

For more information about Azure Data Lake Storage security, see the Microsoft documentation.

Enterprise Data Preparation supports multiuser access to storage locations accessible to HDInsight if Enterprise Security Pack (ESP) is installed in the cluster. In addition, you can use Apache Ranger to configure role-based access control (RBAC) to Hive.

Install Enterprise Data Preparation

Install Enterprise Data Preparation on the Azure HDInsight cluster.

You can create the Enterprise Data Preparation Service and the Interactive Data Preparation Service during the installation process. If you want the installer to create the services, it creates both services on a node.

The installer prompts for cluster configuration associated with the Hadoop environment. The installer also prompts for the Data Preparation repository database connection details.

Informatica recommends that you associate dedicated Model Repository Service and Data Integration Service instances with the Enterprise Data Preparation Service. You can create a Model Repository Service and a Data Integration Service to associate with the Enterprise Data Preparation Service during installation.

When you install Enterprise Data Preparation on multiple nodes, you must install the Enterprise Data Preparation binaries in the Enterprise Data Catalog installation directory on a gateway node before you install the binaries on the other nodes. You must install the Enterprise Data Preparation binaries in the Enterprise Data Catalog installation directory to register the Enterprise Data Preparation Service and the Interactive Data Preparation Service components with the Service Manager for the domain.

For information on installing Enterprise Data Preparation, see Informatica Installation for Enterprise Data Preparation.

Create Hadoop File System Connections to the Data Lake

Use the Administrator tool to create a Hadoop File System connection to each data storage location in the HDInsight cluster.

Create Azure Blob Storage Connections

Use the Administrator tool to create a Hadoop File System to create a connection for each Azure Blob Storage container accessible to the HDInsight cluster.

1. Click the **Connections** tab.

2. Select the domain in the Domain Navigator.
3. Select **Actions** > **New** > **Connection**.
4. Select **Hadoop File System**.
5. Configure the connection properties.

   The following table describes the connection properties:

Property	Description
Name	Name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	The description of the connection. The description cannot exceed 765 characters.
User Name	User name to access the location.
NameNode URI	The URI to access the location. Format the URI as follows: `wasb[s]://<blob storage container name>@<storage account name>.blob.core.windows.net/<path>` Specify the file or directory HDFS path name as the `<path>` value. Because containers in Azure storage are key-value stores, there is no true hierarchical file system. The application interprets the slash character (/) inside a blob key as a directory separator. If you do not specify the `<blob storage container name>` and the `<storage account name>` values, the application uses the default file system.
Cluster Configuration	The name of the cluster configuration associated with the HDInsight cluster.

6. Click **Finish**.

Create Azure Data Lake Storage Connections

Use the Administrator tool to create a Hadoop File System connection to each Azure Data Lake Storage Gen 1 location accessible to the HDInsight cluster.

If Enterprise Data Preparation uses an HDFS connection to the default storage location, you do not need to create a connection to the default storage location.

1. Click the **Connections** tab.
2. Select the domain in the Domain Navigator.
3. Select **Actions** > **New** > **Connection**.
4. Select **Hadoop File System**.
5. Configure the connection properties.

The following table describes the connection properties:

Property	Description
Name	Name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	The description of the connection. The description cannot exceed 765 characters.
User Name	User name to access the storage system.

NameNode URI	The URI to access the Azure Data Lake Storage location.
	Format the URI for the default storage location as follows:
	`adl://mydatalakestore/<cluster root path>`
	Format the URI for each additional storage location as follows:
	`adl:// mydatalakestore.azuredatalakestore.net/<file path>`

Cluster Configuration	The name of the cluster configuration associated with the HDInsight cluster.

6. Click **Finish**.

Create Enterprise Data Catalog Resources

Use the Catalog Administrator to create Azure Storage Blob, HDFS, Azure Data Lake Storage, and Hive resources for the data lake in Enterprise Data Catalog. You then use Enterprise Data Preparation to add the resources to the data lake.

A resource is a repository object that represents an external data source or metadata repository. Scanners attached to a resource extract metadata from the resource and store the metadata in Enterprise Data Catalog.

For more information about creating resources and scanners, see "Creating a Resource" in the Informatica Catalog Administrator Guide.

After you create the resources, use the Enterprise Data Preparation user interface to add the resources to the data lake. You must add at least one Hive resource to the data lake. The Hive resource is required to perform export, import, upload, data preparation, and asset publication operations in Enterprise Data Preparation.

For information about adding resources to the data lake, see the "Managing the Data Lake" chapter in the Informatica Enterprise Data Preparation Administrator Guide.

Create Azure Blob Storage or HDFS Resources

Use Informatica Catalog Administrator to create a Microsoft Azure Blob Storage resource or an HDFS resource for each Azure Blob Storage container in the data lake.

You can create an Azure Blob Storage resource to extract metadata from files stored in an Azure Blob Storage container.

For more information about creating an Azure Blob Storage resource, see the "Microsoft Azure Blob Storage" section in the *Informatica 10.2.2 Catalog Administrator Guide*.

You can also create an HDFS resource to import metadata from files stored in an Azure Blob Storage container. Select **WASB** as the **Storage Type** property value.

For more information about creating an HDFS resource, see the "HDFS Resource Connection Properties" section in the *Informatica 10.2.2 Catalog Administrator Guide*.

Create Azure Data Lake Storage Resources

Use Informatica Catalog Administrator to create an Azure Data Lake Store resource for each ADLS location in the data lake.

Before you create the resource, import the SSL certificates in the following directoryon the node where Enterprise Data Catalog runs:

```
<Informatica installation directory>/java/jre/lib/security/cacerts
```

into the infa_truststore.jks file in the following directory on the node:

```
<Informatica installation directory>/services/shared/security/
```

Create a Hive Scanner

Use Informatica Catalog Administrator to create a Hive resource that Enterprise Data Catalog uses to extract metadata from Hive tables in the data lake.

The Hive resource is required to perform export, import, upload, data preparation, and publish operations in Enterprise Data Preparation.

Observe the following guidelines when you create the Hive resource:

- In the **URL** property on the **General** > **Connection Properties** panel, specify the Fully Qualified Domain Name (FQDN) of the Hive server in the JDBC connection URL.
- If you are using operating system profiles, the Hive user name that you specify as the value for the **User** property must be a Hive superuser. For more information about operating system profiles, see the "Operating System Profiles" section in the Informatica Enterprise Data Preparation Administrator Guide.
- Import the relevant connectors to extract metadata from Hive sources.

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